

I251/I252/I261 Cosmetic Spec

IT ME, A-DEV

SAMSUNG CONFIDENTIAL

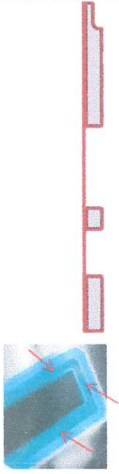
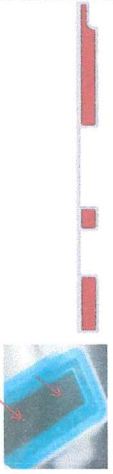
OLED

IP Model Cover Material Cosmetic Spec

Cover Material

A-DEV IT ME Part

SDC IT Model Cover Cosmetic Spec

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Fiber Burr	Vision area	* Vision area : Material Side area + 15mm area 	Cover-PCB	Size	Visual	AQL .65
	Non-Vision Area					
PSA Burr	Max Height 0.2mm		All Cover (Cover-COF exception)	Size	Visual	AQL .65
Tear	Max 0.5mm		All Cover (Cover-COF exception)	Size	Visual	AQL .65
Stain	Allowed (only non-removable & no Z-impact)		All Cover (Cover-COF exception)	Size	Visual	AQL .65
Particle	$\phi \leq 0.2\text{mm}$ Ignore ; $\phi > 0.2\text{mm}$ 3ea ; $\phi \geq 0.3\text{mm}$ 0ea		All Cover (Cover-COF, Cover-PCB exception)	Size	Visual	AQL .65
	$\phi \leq 0.2\text{mm}$ Ignore ; $\phi > 0.2\text{mm}$ 3ea ; $\phi \geq 0.3\text{mm}$ 0ea (Only RED area spec)					
Particle	$\phi \leq 1.0\text{mm}$ Ignore ; $1.0\text{mm} < \phi \leq 1.2\text{mm}$ 2ea ; $1.2\text{mm} < \phi$ 0ea (Only RED area spec)		Cover-PCB	Size	Visual	AQL .65
						

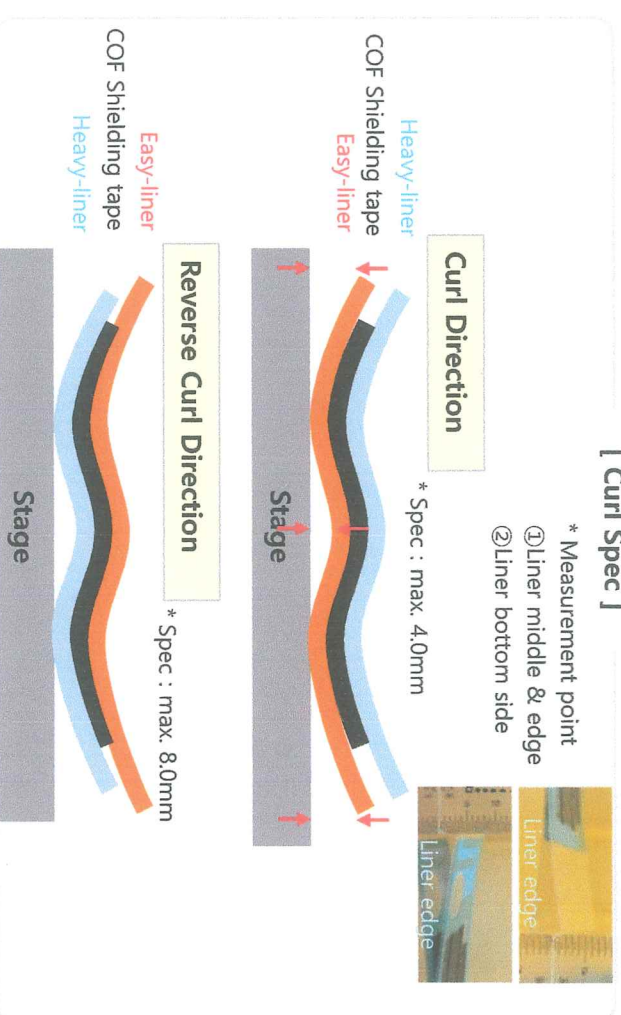
SDC IT Model Cover Cosmetic Spec

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Deformation	Not Allowed		All Cover	Size	Visual	AQL .65
Kiss Cut (Main materials)	Now Allowed @ Heavy-liner / Easy liner (Cover-window : Allowed @ Easy-liner)		All Cover	Size	Visual	AQL .65
Discoloration	Ignore @ PSA side (if non-removable & no Z-impact)		All Cover (Cover-COF exception) Cover PCB	Size	Visual	AQL .65
Compressed Felt	Ignore @ PSA side (if non-removable & no Z-impact)		All Cover (Cover-COF exception) All Liner/Filler/Blocker	Size	Visual	AQL .65
PSA Overflow	Max 0.1mm from edge		All Cover (Cover-COF exception) All Liner/Filler/Blocker	Size	Visual	AQL .65

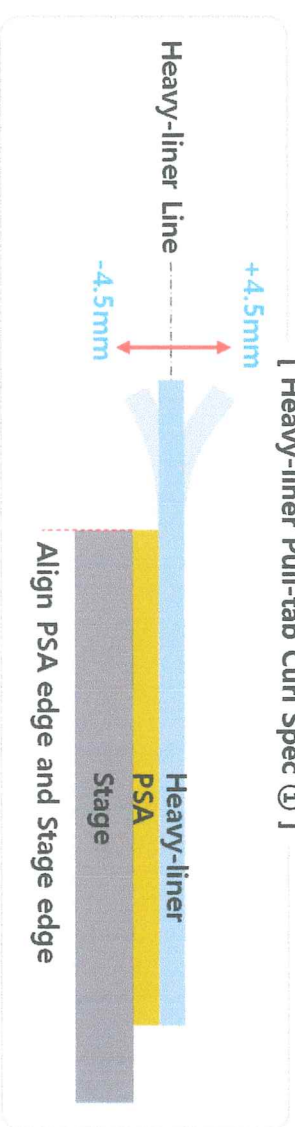
SDC IT Model Cover Cosmetic Spec

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Curl	Max 6.0mm (Both Curl & Reverse Curl) Curl : Max 4.0mm Reverse Curl : Max 8.0mm		All Cover (Cover-COF exception)	Size	Visual	AQL .65
Curl			Cover-COF	Size	Visual	AQL .65
Heavy-liner Pull-tab Curl ①	±4.5mm from Heavy-liner position (refer below Measurement SOP)	Only Heavy-liner (Easy-liner exception)	All Cover (Cover-PCB exception)	Value	Visual	AQL .65
Heavy-liner Pull-tab Curl ②	+2/-7.5mm from Heavy-liner position (refer below Measurement SOP)	Only Heavy-liner (Easy-liner exception)	Cover-PCB	Value	Visual	AQL .65

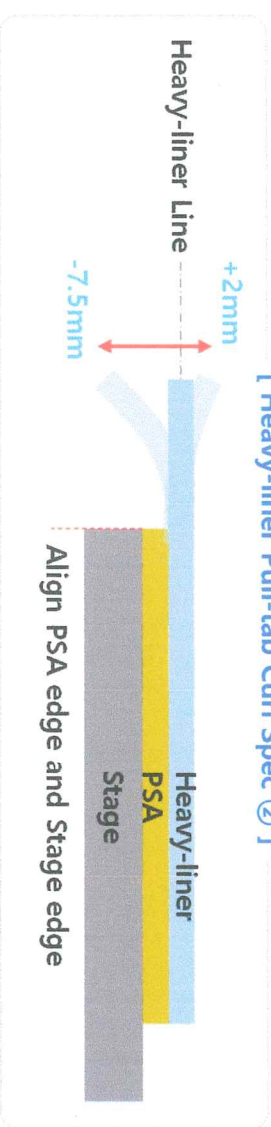
[Curl Spec]



[Heavy-liner Pull-tab Curl Spec ①]



[Heavy-liner Pull-tab Curl Spec ②]

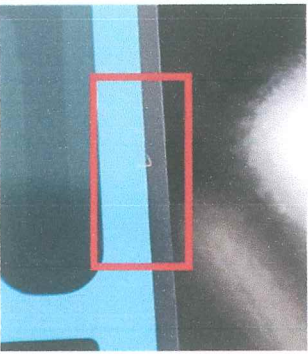
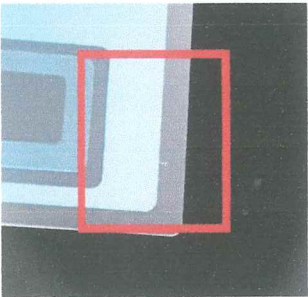
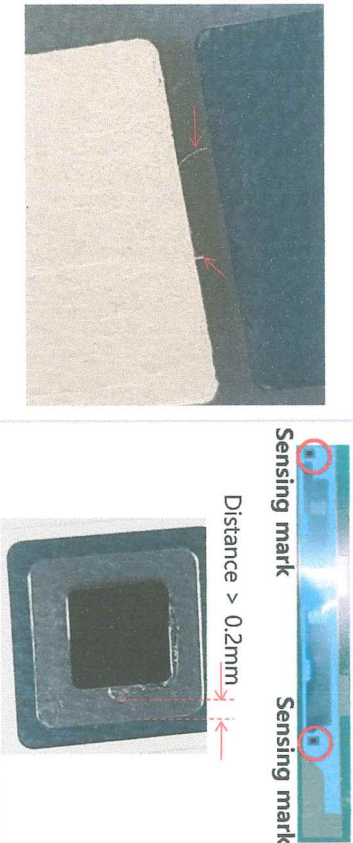
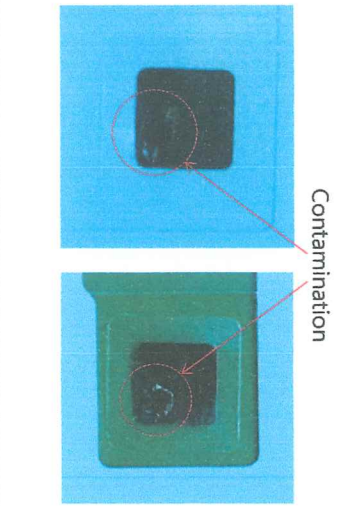


SDC IT Model Cover Cosmetic Spec

Defects	Criteria		Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Bubble	Heavy-liner	[Circle Type] $\phi \leq 0.65\text{mm}$ Ignore $0.65\text{mm} < \phi \leq 2\text{mm}$ 4ea, 2mm $< \phi \leq 4\text{mm}$ 2ea, $\phi > 4\text{mm}$ 0ea [Liner Type] Width $\leq 3\text{mm}$ Ignore	Circle type bubble inner liner type bubble → Ignore	All Cover (Cover-COF exception)	Size	Visual	AQL .65
	Easy liner	Ignore					
Pin Hole	Heavy-liner	[Circle Type] $\phi \leq 0.65\text{mm}$ Ignore $0.65\text{mm} < \phi \leq 2\text{mm}$ 4ea, 2mm $< \phi \leq 4\text{mm}$ 2ea, $\phi > 4\text{mm}$ 0ea [Liner Type] Width $\leq 3\text{mm}$ Ignore		All Cover (Cover-COF/Cover-PCB exception)	Size	Visual	AQL .65
	Easy liner	Ignore					
Pin Hole	Heavy-liner	[Circle Type] $\phi \leq 1\text{mm}$ Ignore $1\text{mm} < \phi \leq 2\text{mm}$ 4ea, 2mm $< \phi \leq 4\text{mm}$ 2ea, $\phi > 4\text{mm}$ 0ea [Liner Type] Width $\leq 3\text{mm}$ Ignore		Cover-PCB	Size	Visual	AQL .65
	Easy liner	Ignore					

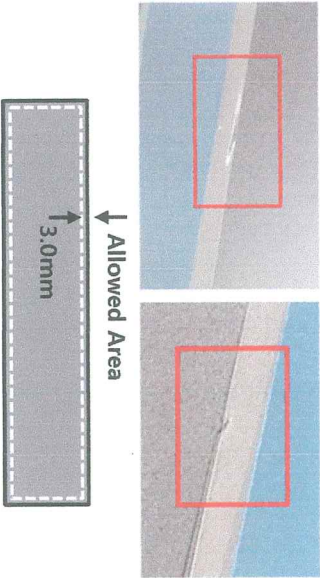
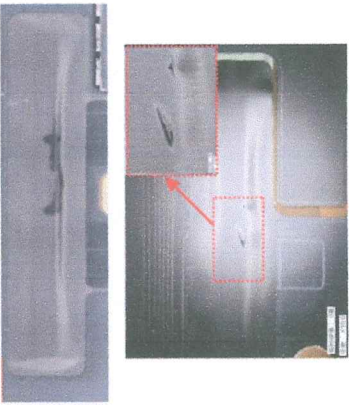
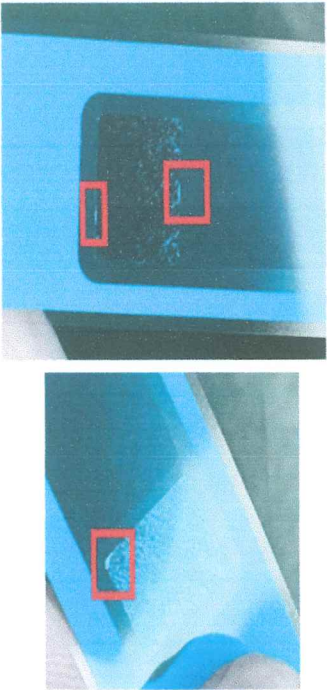
SDC IT Model Cover Cosmetic Spec

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Edge Fiber	Non-removable : Max Height 0.45mm , Qty = unlimited Removable : OK	Image#1	All Liner/Filler/Blocker All Blocker on Material	Size	Visual	AQL .65
Magnetic Burr	Not allowed	Image#2	Cover-FPC	Size	Visual	AQL .65
Sensing mark PSA residue	"Blocker Edge to PSA residue edge" distance > 0.2mm , allowed	Image#3	Cover-FPC	Size	Visual	AQL .65
Sensing mark Contamination	Not allowed	Image#4	Cover-FPC	Size	Visual	AQL .65

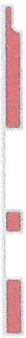
Image#1	Image#2	Image#3	Image#4
			

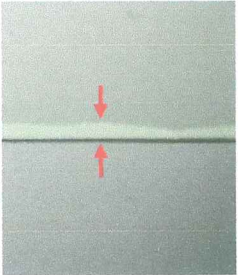
SDC IT Model Cover Cosmetic Spec

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Easy liner Deformation	Allowed 3.0mm area from Easy liner edge (Not allowed if deformation @ guide hole)	Image #1 Only apply until Display Build (JUNE 23Y) Due to manual attachment process	All Cover (Sheet Type)	Size	Visual	AQL .65
Press mark	Not Allowed	Image #2	All Cover (Sheet Type)	Size	Visual	AQL .65
Glint / Wrinkle	Allowed	Image #3	Cover FPC	Size	Visual	AQL .65

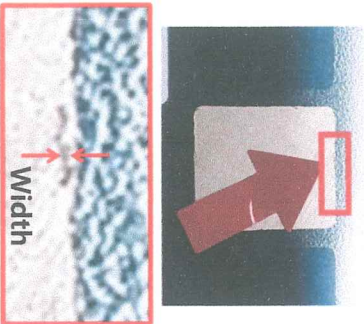
Image#1	Image#2	Image#3
		

SDC IT Model Cover Cosmetic Spec

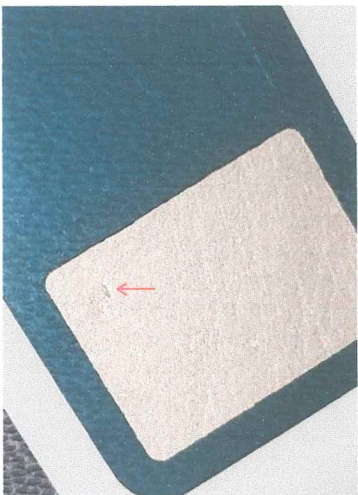
Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Scratch	Length ≤ 20mm, Ignore Length > 20mm, Qty 3ea		All Cover (Cover-COF, Cover-PCB exception)	Size	Visual	AQL .65
Scratch	Length ≤ 20mm, Ignore Length > 20mm, Qty 3ea	 	Cover-PCB	Size	Visual	AQL .65
Scratch	Allowed (if no Thickness problem & Z-impact)	 	Cover-PCB	Size	Visual	AQL .65
Edge Linear Bubble	Max Width 1.0mm from edge (all around)	Image#1	Cover Window	Size	Visual	AQL .65
Magnetic Crack	Width ≤ 0.15mm , ignore 0.15 < Width ≤ 0.2mm , 3ea (No PSA missing around the crack)	Image#2	Cover FPC	Size	Visual	AQL .65
Magnetic Dent	Allowed (No Penetration)	Image#3	Cover FPC	Size	Visual	AQL .65



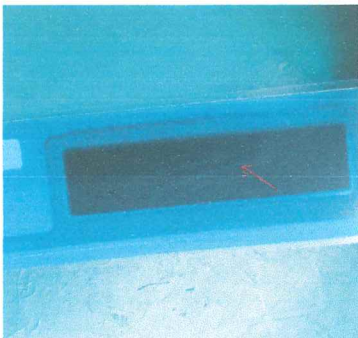
Image#1



Image#2



Image#3



Image#4

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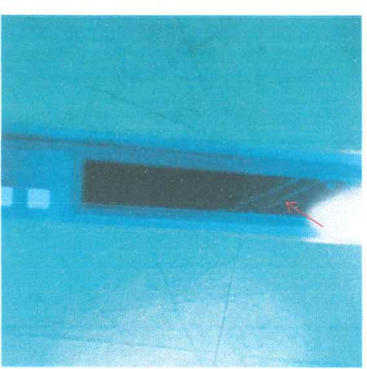
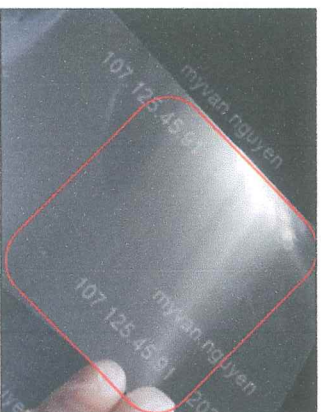
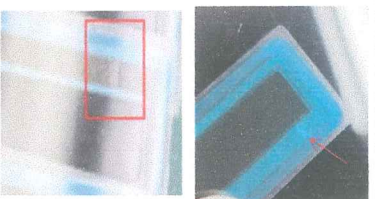
MCE / A-DEV

SAMSUNG DISPLAY

SDC IT Model Cover Cosmetic Spec

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Impressions Indent	Length $\leq$ 20mm, Ignore, Length $>$ 20mm, Qty 3ea (if non-removable & no Z-impact)	Image#1 Image#2	All Cover (Cover-COF exception)	Size	Visual	AQL .65
Imprints	Length $\leq$ 20mm, Ignore, Length $>$ 20mm, Qty 3ea (if non-removable & no Z-impact)	Image#1	All Cover (Cover-COF, Cover-PCB exception)	Size	Visual	AQL .65
Imprints	Length $\leq$ 20mm, Ignore, Length $>$ 20mm, Qty 3ea (if non-removable & no Z-impact)	Image#2 Image#4	Cover-PCB	Size	Visual	AQL .65
Imprints	Allowed (if no Thickness problem & Z-impact)					
Stripes	Allowed (No Stripes after lamination on Flat-Glass surface)	Image#3	Cover window	Size	Visual	AQL .65

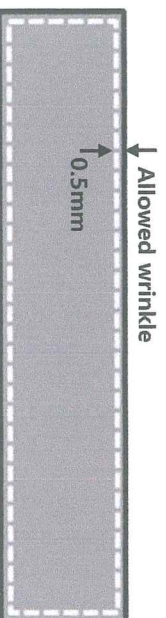
Impressions caused by heavy liner cut out



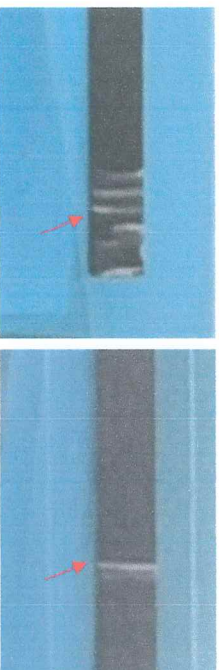
SDC IT Model Cover Cosmetic Spec ... Wrinkle

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Wrinkle	Not Allowed except for specific area / material (Refer below note of each cover material)			Size	Visual	AQL .65
Guide#1	Allowed (Only Inner 0.5mm area from foam edge)	Image #1	Cover Spacer	Size	Visual	AQL .65
Guide#2	Not allowed (Inspect on Easy-liner)	Image #2	Cover Spacer	Size	Visual	AQL .65
Guide#3	Allowed (only BLK PSA)	Image #3	Cover PCB	Size	Visual	AQL .65
Guide#4	No wrinkle after lamination on Flat-Glass surface → Allowed		Cover window	Size	Visual	AQL .65

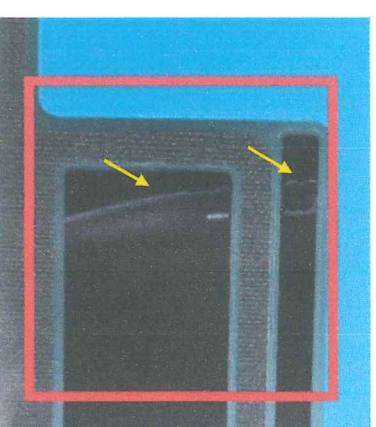
Image#1



Image#2

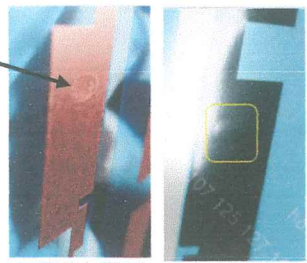
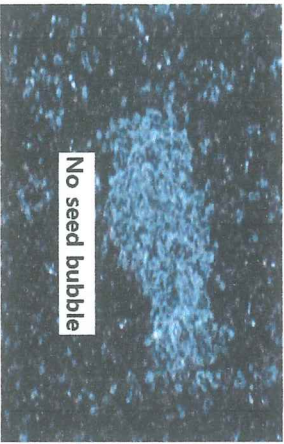
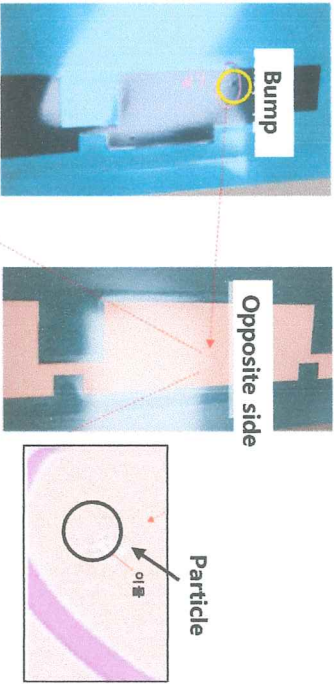
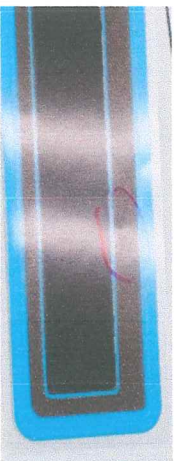


Image#3

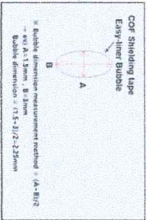
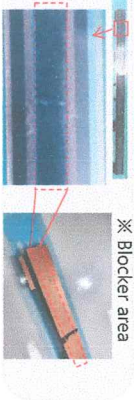


SDC IT Model Cover Cosmetic Spec ... Dent / Bump

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Dent	$\phi \leq 1.0\text{mm} : 5\text{ea}$ $1.0\text{mm} < \phi \leq 1.5\text{mm} : 3\text{ea}$ $1.5\text{mm} < \phi \leq 2.0\text{mm} : 1\text{ea}$		All Cover (Cover-COF exception)	Size	Visual	AQL .65
Bump	$\phi \leq 1.0\text{mm} : 5\text{ea}$ $1.0\text{mm} < \phi \leq 1.5\text{mm} : 3\text{ea}$ $1.5\text{mm} < \phi \leq 2.0\text{mm} : 1\text{ea}$		All Cover (Cover-COF, Cover-PCB exception)	Size	Visual	AQL .65
Bump	$\phi \leq 2.5\text{mm} : 1\text{gnore}$ $2.5\text{mm} < \phi \leq 3.0\text{mm} : 5\text{ea}$ $3.0\text{mm} < \phi \leq 3.5\text{mm} : 3\text{ea}$	Image#4	Cover-PCB	Size	Visual	AQL .65
Guide#1	Bump / Dent phenomenon caused by bubble on opposite side $\rightarrow$ Ignore	Image#1		Size	Visual	AQL .65
Guide#2	No seed center of bubble of heavy liner $\rightarrow$ Bubble	Image#2		Size	Visual	AQL .65
Guide#3	Bump / Dent caused by particle on opposite side $\rightarrow$ Bump / Dent	Image#3		Size	Visual	AQL .65

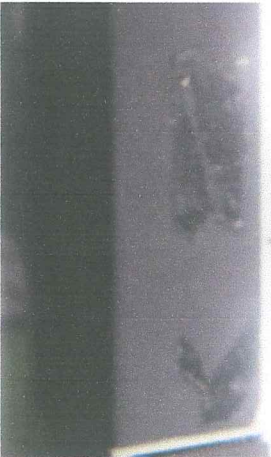
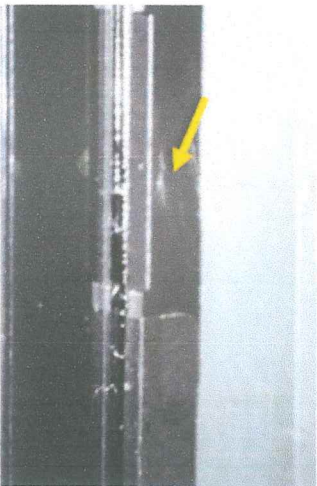
Image#1	Image#2	Image#3	Image#4
 Bubble btw liner & PSA	 No seed bubble	 Bump Opposite side Particle	

SDC IT Model Cover Cosmetic Spec ... Cover-COF

Defects	Criteria		Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Bubble	Heavy-liner	$\Phi > 5\text{mm}$, not allowed $\Phi \leq 5\text{mm}$, allowed	 	Cover COF	Size	Visual	AQL .65
	Easy-liner	[Blocker area]: $\Phi > 5\text{mm}$, not allowed $\Phi \leq 5\text{mm}$, allowed [Others]: Ignore					
Tear	Not allowed			Cover COF	Size	Visual	AQL .65
Stain	Allowed (only non-removable & no Z-impact)			Cover COF	Size	Visual	AQL .65
Discoloration	Allowed (No AG or Cu exposure)			Cover COF	Size	Visual	AQL .65
Scratch	Scratch without Copper exposure is ok			Cover COF	Size	Visual	AQL .65
Edge Fiber	Not allowed			Cover COF (Liner/Filter/Blocker exception)	Size	Visual	AQL .65

SDC IT Model Cover Cosmetic Spec ... Cover-COF

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Wrap / Broken	Not allowed	Image #1	Cover-COF	Size	Visual	AQL .65
Glue residue	Not allowed	Image #2	Cover-COF	Size	Visual	AQL .65
Color fade	Color fade is acceptable without Ag or Copper exposure	Image #3	Cover-COF	Size	Visual	AQL .65
Copper Exposure	Not Allowed		Cover COF	Size	Visual	AQL .65
Tear	Not Allowed		Cover COF	Size	Visual	AQL .65

Image#1	Image#2	Image#3
		

SDC IT Model Cover Cosmetic Spec ... Cover-COF

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling				
Impressions Imprints Indent	Length ≤ 20mm, Ignore Length > 20mm, Q'ty 3ea (if non-removable & no Z-impact)	Image#1 Image#2 Image#3 Image#4	Cover-COF	Size	Visual	AQL .65				
Particle (Removable)	Φ < 0.15 mm ignore ; Φ≥ 0.15mm 3ea ; Φ≥ 0.30mm 0ea ; 단, 직경10mm 내 군집 없을 것, 이물 제거 후 전수 사용 가능	Between COF shielding tape and Heavy-liner	Cover-COF	Size	Visual	AQL .65				
Particle (Non-Removable)	0 < Φ ≤ 1.5mm , 5ea : Allowed 1.5mm < Φ : Not allowed	Between COF shielding tape and Easy-liner	Cover-COF	Size	Visual	AQL .65				
PSA Burr	<table><tr><td>Vision area</td><td>Not Allowed</td></tr><tr><td>Non-Vision Area</td><td>Max Height 0.2mm</td></tr></table>	Vision area	Not Allowed	Non-Vision Area	Max Height 0.2mm	Image #3	Cover COF	Size	Visual	AQL .65
Vision area	Not Allowed									
Non-Vision Area	Max Height 0.2mm									

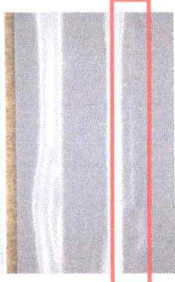
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Image#2

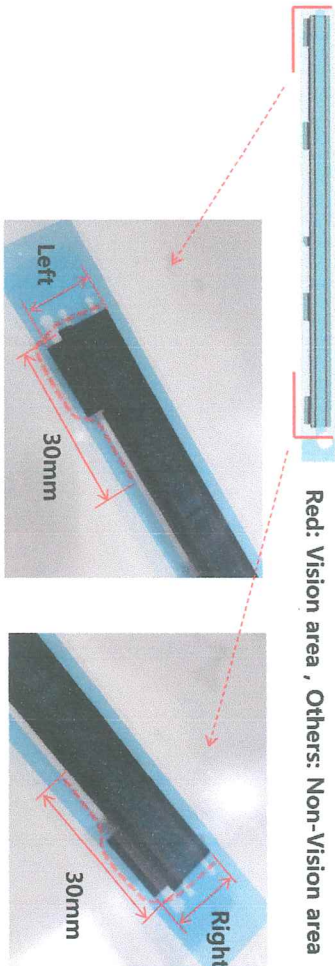
Image#3



Phenomenon#1



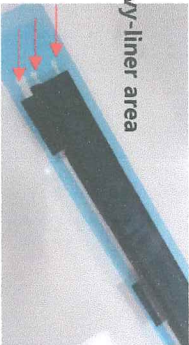
Phenomenon#2



Red: Vision area , Others: Non-Vision area

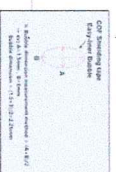
SDC IT Model Cover Cosmetic Spec ... Cover-COF

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Wrinkle	Dimension: Width ≤ 0.8mm , Length ≤ 8mm	Image #1	Cover-COF	Size	Visual	AQL .65
Guide#1	Allowed (Only Non Heavy-liner area)	Image #2 Image #3 Image #4	Cover COF	Size	Visual	AQL .65

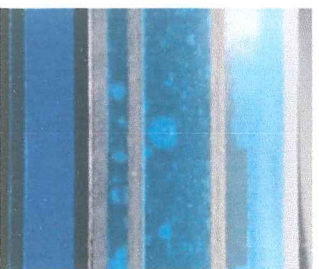
Image#1	Image#2	Image#3	Image#4
			

SDC IT Model Cover Cosmetic Spec ... Cover-COF

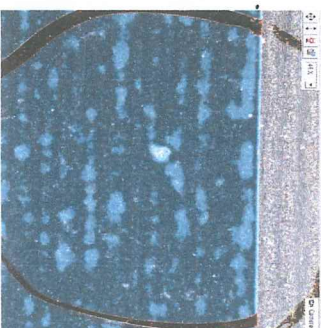
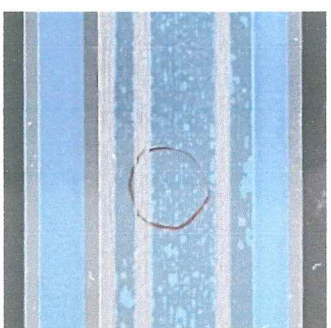
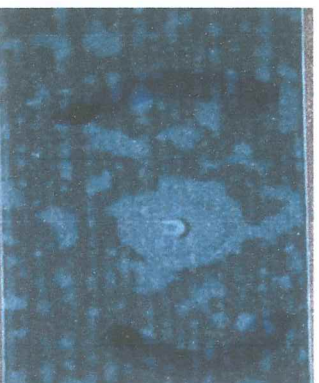
Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Dent / Pin-hole &	$\Phi \leq 2.2\text{mm}$, Allowed $2.2\text{mm} < \Phi$, Not allowed	Follow Samju inspection Method	Cover-COF	Size	Visual	AQL .65
Bump	※ Exposure of Cu or adhesive : Not allowed					
Guide#1	※ Dent/Bump Calculation method Dent image (No exposed Cu)	Image#1				
Guide#2	Dent image (Exposed Cu)	Image#2				
Guide#3	Bump image	Image#3				



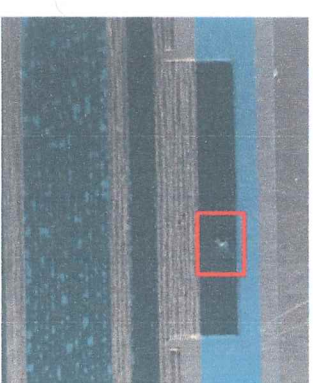
Image#1



Image#2



Image#3



SDC IT Model Cover Cosmetic Spec ... Cover-PCB

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Flatness / warpage	Flatness Spec : Max. - Min. ≤ 0.3	Only for Aluminium material of I261	Cover-PCB	Size	OMM	AQL .65
Guide#1 (for flatness / warpage)	Measurement SOP	Image#1				
Bright edge	Allowed (if no Thickness & MCO problem and Z-impact)	Image#2	Cover-PCB	Size	Visual	AQL .65

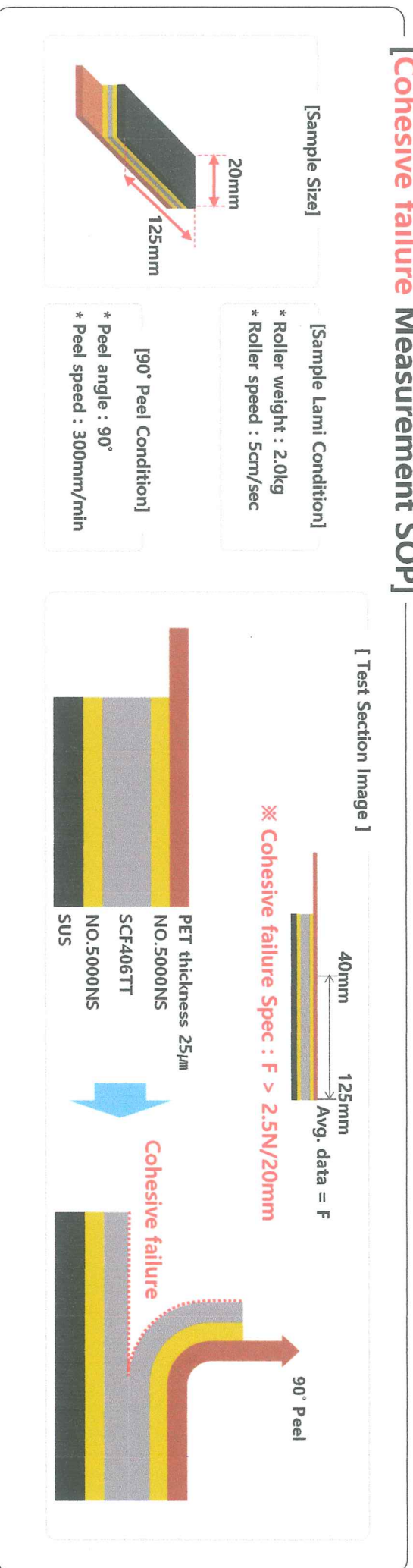
Image#1

Image#2

SDC IT Model Cover Cosmetic Spec ... Cover-Spacer

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Foam Layer Cohesive Failure	$F > 2.5\text{N}/20\text{mm}$ ($\approx 255\text{gf}/20\text{mm}$) ※ Calculate the result "F" as the average value of the 40mm to 125mm area ※ Refer below image	Follow Below SOP	Cover-spacer	Value	UTM	AQL .65

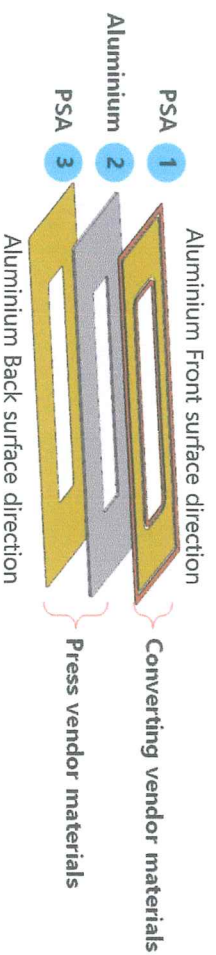
[Cohesive failure Measurement SOP]



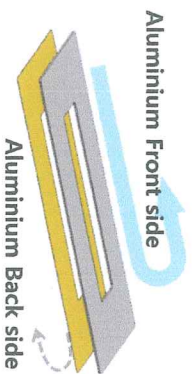
SDC IT Model Cover Cosmetic Spec ... Cover-PCB

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Peeling force of Aluminium surface	[Front side] A area : $F \geq 800$ gf B area : $F \geq 500$ gf [Back side] A area : $F \geq 1100$ gf B area : $F \geq 800$ gf	Follow Below SOP	Cover-PCB of 1261	Value	UTM	AQL .65

Materials Structure of Cover-PCB

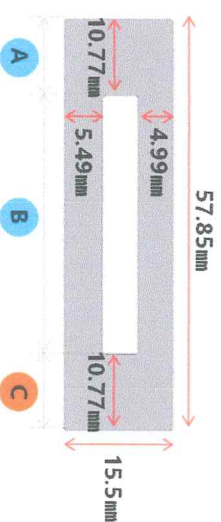


Measurement Condition

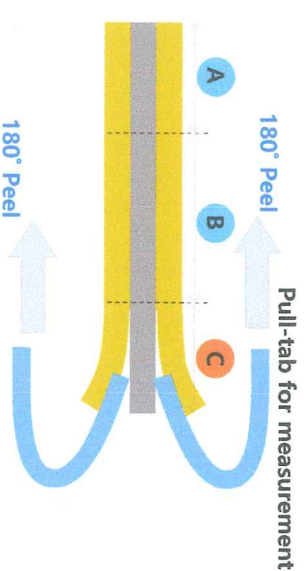


- * Measurement Surface : Front side / Back side
- * PSA for Front side peeling : TESA7475
- * PSA for Back side peeling : PA0080H-70 (PSA from Press vendor)
- * Measure the Avg. Peeling-force value for A and B area
- Area C is used for Pull-tab area, No need to measure Peeling force

Aluminium Outline



- A : Measurement area
- B : Measurement area
- C : Pull-tab area for measurement

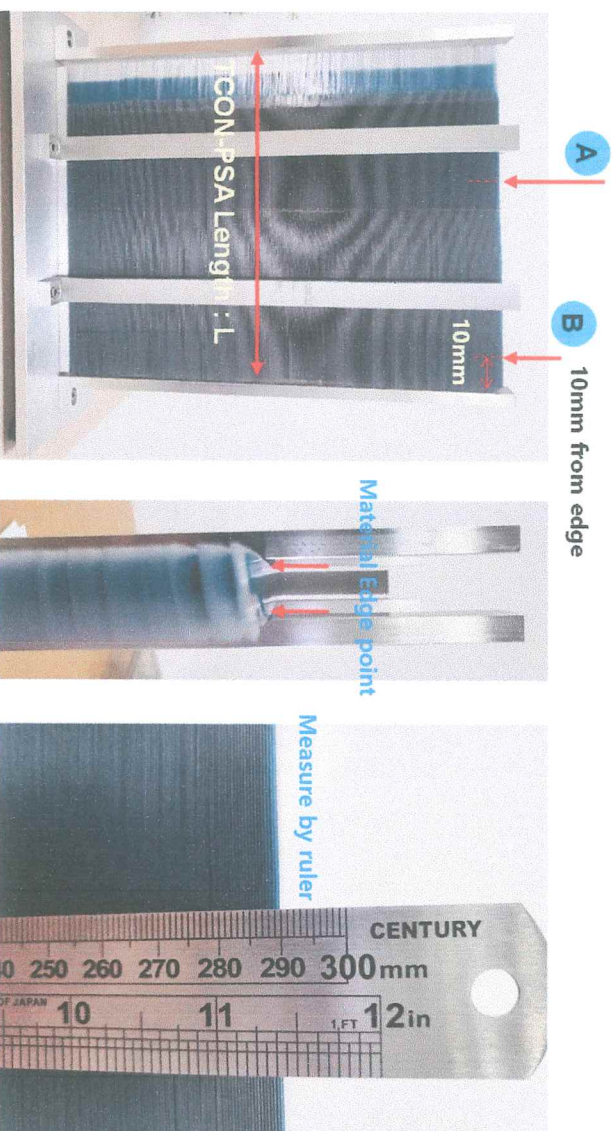


SDC IT Model Cover Cosmetic Spec ... Cover-PCB

Defects	Criteria	Remark	Material Of Application	Judgement Priority	Inspection Method	Sampling
Magazine Tilt	[Tilt Spec #1] ① and ② Gap ≤ 3.5 mm ③ and ④ Gap ≤ 3.5 mm [Tilt Spec #2] min. of A and min. of B Gap ≤ 4.5 mm	Follow Below SOP ※ Sample Stack Qty : 500pcs ※ If the Qty is less than 500ea, it needs to collect other materials and make a Qty to 500ea, and then put the materials to SDV process after evaluation	Cover-PCB of I251/252	Value	Visual	AQL .65

[Measurement Area Definition]

$\frac{1}{3}$ X L area from H/L Pull-tab edge



[Measurement Value]

* Measurement : ①,②,③,④ Height of **A** / **B** area

※ min. of **A** : min. height of ① and ②
 min. of **B** : min. height of ③ and ④

